

SKILL - 7

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[3]: import pandas as pd

df = pd.read_csv("placement_predict_50k Dataset (2).csv")

print(df.head())
print(df.columns)
```

	StudentID	Gender	City	CollegeTier	Stream	Specialisation	Hostel	\
0	1	Male	Ahmedabad	Tier2	ECE	Networking	No	
1	2	Female	Mumbai	Tier2	ECE	DataScience	Yes	
2	3	Male	Kolkata	Tier2	IT	DataScience	Yes	
3	4	Male	Jaipur	Tier1	CS	AI	No	
4	5	Male	Pune	Tier2	IT	DataScience	Yes	

	HistoryOfBacklogs	SGPA_Sem1	SGPA_Sem2	...	Publications	\
0	No	6.02	6.54	...	0	
1	Yes	5.84	5.12	...	0	
2	No	4.91	5.29	...	0	
3	No	7.67	8.03	...	0	
4	No	8.14	8.97	...	1	

	AptitudeTestScore	SoftSkillsRating	CodingTestScore	MockInterviewScore	\
0	66.7	2.2	49.4	47.8	
1	48.2	2.4	26.7	25.8	
2	73.8	2.8	67.7	41.5	
3	69.8	2.7	66.9	48.0	
4	73.1	2.1	71.7	61.7	

	ExtraCurricular	CGPA_Tier	PlacementStatus	IsAnomaly	Salary	Package
0	0	Low	0	0	0.00	
1	0	Low	0	0	0.00	
2	0	Low	1	0	3.89	
3	0	Mid	1	0	8.37	
4	1	High	1	0	18.99	

```
[5 rows x 32 columns]
Index(['StudentID', 'Gender', 'City', 'CollegeTier', 'Stream',
      'Specialisation', 'Hostel', 'HistoryOfBacklogs', 'SGPA_Sem1',
      'SGPA_Sem2', 'SGPA_Sem3', 'SGPA_Sem4', 'SGPA_Sem5', 'SGPA_Sem6',
      'SGPA_Sem7', 'SGPA_Sem8', 'CGPA', 'AttendancePercent', 'Internships',
      'Projects', 'Workshops', 'Certifications', 'Publications',
      'AptitudeTestScore', 'SoftSkillsRating', 'CodingTestScore',
      'MockInterviewScore', 'ExtraCurricular', 'CGPA_Tier', 'PlacementStatus',
      'IsAnomaly', 'Salary', 'Package'],
      dtype='object')
```

[5]:

```
#BIVARIATE ANALYSIS

#Scatter Plot
plt.figure(figsize=(6,4))
sns.scatterplot(
    data=df,
    x="CGPA",
    y="AttendancePercent"
)
plt.title("CGPA vs Attendance Percentage")
plt.show()

#Regression Plot
plt.figure(figsize=(6,4))
sns.regplot(
    data=df,
    x="CGPA",
    y="CodingTestScore"
)
plt.title("CGPA vs Coding Test Score")
plt.show()

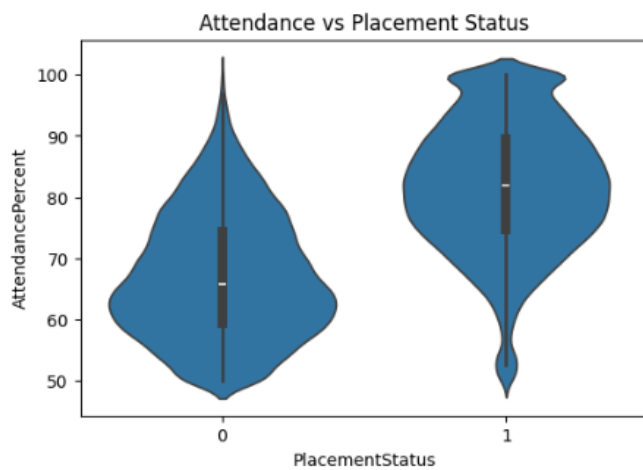
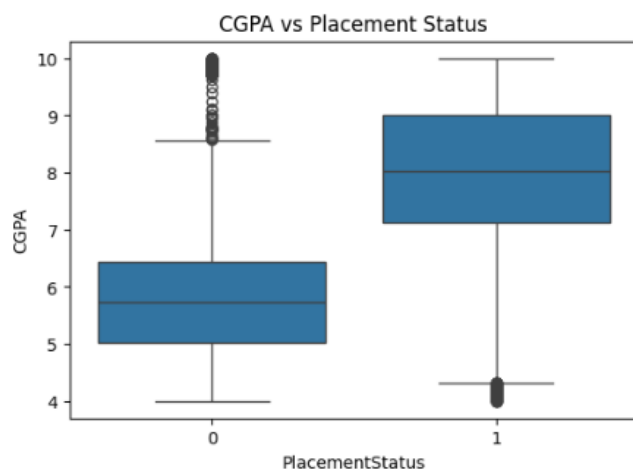
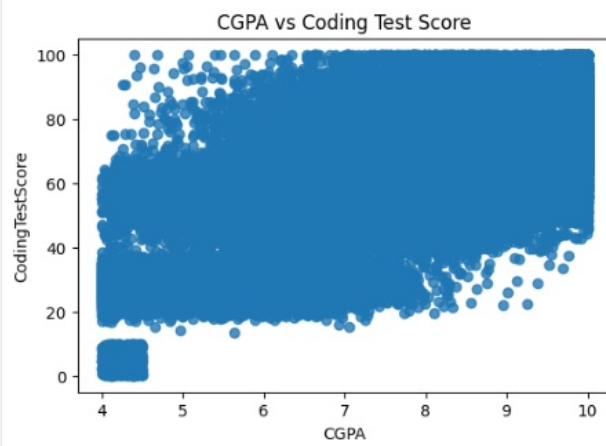
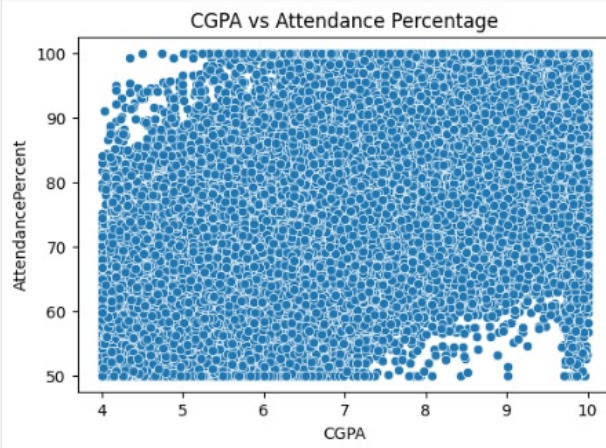
#Box Plot
plt.figure(figsize=(6,4))
sns.boxplot(
    data=df,
    x="PlacementStatus",
    y="CGPA"
)
plt.title("CGPA vs Placement Status")
plt.show()

#Violin Plot
plt.figure(figsize=(6,4))
sns.violinplot(
    data=df,
    x="PlacementStatus",
    y="AttendancePercent"
)
plt.title("Attendance vs Placement Status")
plt.show()

#Gender vs Placement
plt.figure(figsize=(6,4))
sns.countplot(
    data=df,
    x="Gender",
    hue="PlacementStatus"
)
plt.title("Gender vs Placement Status")
plt.show()

#Stream vs Placement
plt.figure(figsize=(10,5))
sns.countplot(
    data=df,
    x="Stream",
```

```
plt.show()
```



```
'MockInterviewScore', 'ExtraCurricular', 'CGPA_Tier', 'PlacementStatus',  
'IsAnomaly', 'Salary Package'],  
dtype='object')
```

```
[4]: import matplotlib.pyplot as plt  
import seaborn as sns  
  
plt.figure(figsize=(6,4))  
sns.countplot(data=df, x="PlacementStatus")  
plt.title("Placement Status Distribution")  
plt.show()
```

